

Problem

Rather than applying the shortcut technique used in Section 7.6, use KVL to obtain Eq. (7.60).

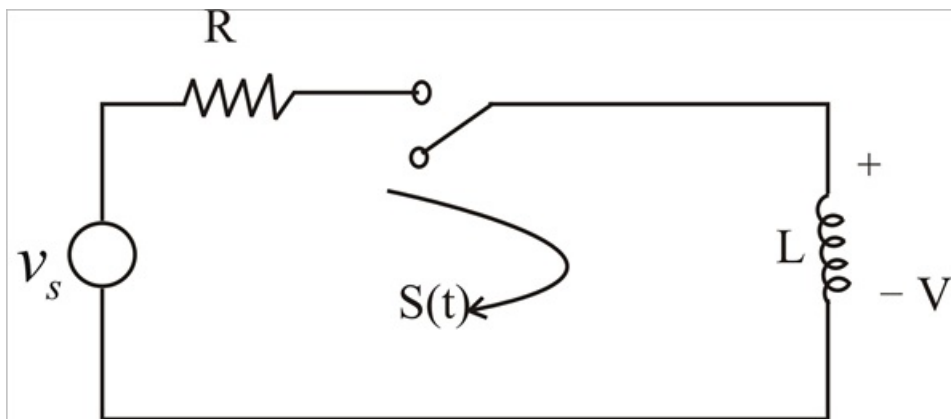
Eq. (7.60):

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-t/\tau}$$

Step-by-step solution

Step 1 of 3

Consider circuit below:-



Step 2 of 3

After the switch is closed, applying KVL gives,

$$V_s = R_1 + L \frac{di}{dt}$$

$$L \frac{di}{dt} = -r \left(i - \frac{V_s}{R} \right)$$

$$\frac{di}{1 - \frac{V_s}{R}} = \frac{-R}{L} dt$$

Integrating both sides,

$$\ln \left(i - \frac{V_s}{R} \right) \Big|_{I_0}^{i(t)} = \frac{-R}{L} t .$$

Step 3 of 3

$$\ln \left(\frac{i - \frac{V_s}{R}}{I_0 - \frac{V_s}{R}} \right) = \frac{-t}{\tau}$$

$$\text{Or, } \left(\frac{i - \frac{V_s}{R}}{I_0 - \frac{V_s}{R}} \right) = e^{\frac{-t}{\tau}}$$

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{\frac{-t}{\tau}}$$

Same as in the equation 7.60.